



TEMASEK JUNIOR COLLEGE, SINGAPORE
Year Two
Preliminary Examinations 2008
Higher 2

COMPUTING

9754/01

Paper 1

9th September 2008

Additional Materials: Answer Paper

3 hours

READ THESE INSTRUCTIONS FIRST

Write your *Civics Group* and *Name* on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs, music or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **5** printed pages and 1 blank page.



TEMASEK JUNIOR COLLEGE, SINGAPORE
PASSION PURPOSE DRIVE



Answer **all** questions

- 1
- (a)
- (i) Explain what are *registers*. [2]
- (ii) Explain **two** differences between registers and cache memory. [4]
- (iii) Describe how two numbers are added during the fetch-execute cycle using the CPU registers. [6]
- (b) Embedded systems can be found in all aspects of our life.
- (i) What is an *embedded system*? [1]
- (ii) Give **two** examples where embedded systems can be found. [2]
- (iii) Explain why an embedded system is needed in that two examples you have given in (ii). [2]
- (c) Tablet PCs are an alternative to standard notebook PCs.
- (i) Briefly describe how a tablet PC functions. [3]
- (ii) Give a scenario where tablet PCs are required. Explain why. [2]
- 2
- (a) When a function is invoked, the data in the parentheses is passed into the receiving function using either *pass by value* or *pass by reference* or sometimes both.
- (i) Explain, with suitable examples, the terms *pass by value* and *pass by reference*. [4]
- (ii) Give a situation where by pass by reference is used over pass by value. Explain why. [3]
- (b) Recursive functions are sometimes used by programmers in their codes.
- (i) Describe the key features that make a function recursive. [2]
- (ii) Describe **two** advantages of using a recursive function. [2]
- (c)
- (i) There are many sorting algorithms available. What considerations would a programmer take into when choosing a suitable sorting algorithm. Explain your answer. [4]
- (ii) The quicksort is one of the fastest sorting algorithm you have encountered.
- Show how quicksort can be used to sort the following numbers below into ascending order.
- 230 112 413 221 103 504 [5]

- 3 (a) Distinguish between imperative and declarative programming. [4]
- (b) Abstraction is a powerful concept that is recurring in the world of computing.
- (i) Explain what is meant by abstraction in programming. [2]
- (ii) Give an example of how the concept of abstraction is achieved in imperative programming. [2]
- (c) Using suitable examples, explain the following concepts in Object-Oriented Programming.
- (i) Object
- (ii) Encapsulation
- (iii) Inheritance [6]
- (d) Compilers and interpreters are used to convert source codes into working programs.
- (i) Briefly describe the compilation process. [5]
- (ii) Explain why an interpreter is often used to test a program during its development stage. [2]
- 4 Students in XYZ Junior College use a Learning Management System (LMS) on the school intranet to retrieve subject materials such as lecture notes and tutorial questions. They interact with their tutor on the LMS when necessary and also submit their assignments online.
- (a) Explain what an *intranet* is and how it differs from the *internet*. [3]
- (b) Describe **two** advantages for the school through the deployment of the LMS. [4]
- (c) Suppose that the school would like to include video streaming for on-line lecture.
- (i) Describe **two** hardware that would be necessary to make this a possibility. [4]
- (ii) Briefly describe how digital video signals can be transferred across cyberspace to reach students viewing the streaming video from home. [2]
- (d) Explain why the LMS should be deployed within an intranet. [3]
- (e) The school's system may be susceptible to denial of service attacks.
- (i) Explain what is meant by *denial of service* attacks. [2]
- (ii) Give **two** ways in which such attacks can be reduced. [2]

- 5 **(a)** One of the main role of an operating system (OS) is that of a device manager.
- (i)** Briefly state what the device manager needs to do in order to ensure a smooth and efficient device management system. [3]
- (ii)** In order to facilitate communications between devices, the OS makes use of buffers and direct memory access (DMA).
- Explain the terms *buffer* and *direct memory access*. [4]
- (b)** Real time systems can be found in traffic monitoring services.
- (i)** State **two** characteristics of the real-time control operating systems. [2]
- (ii)** Explain how the two characteristics mentioned in **(i)**, help in traffic monitoring. [2]
- 6 **(a)** Indexed-Sequential files are one of the many file organisation methods used.
- (i)** Explain how indexed-sequential files approach random file access. [4]
- (ii)** Explain what multi-level indexes are and why they are needed. [4]
- (iii)** Give **one** example where indexed sequential files are used. [1]
- (b)** **(i)** What is a data dictionary? [1]
- (ii)** List **three** items that can be found in a data dictionary. [3]
- (c)** By using a suitable example, explain why controlled data redundancy is still required in databases. [3]

- 7 Two one-dimensional arrays and their contents are shown below.

A[1]	A[2]	A[3]	A[4]	A[5]
3	4	4	1	3

C[1]	C[2]	C[3]	C[4]
1	0	2	2

Array A contains numbers ranging from 1 to 4 and array B contains the frequency with which each number appears in array A. Eg C[1] means that the number 1 appears once in array A.

- (i) Consider the following algorithm.

```

for i = 2 to 4
    C[i] = C[i] + C[i - 1]
endfor

```

List down the new values within the cells of array C.

[3]

- (ii) Using the new values in array C, consider the next algorithm.

```

for j = length(A) to 1
    B[C[A[j]]] = A[j]
    C[A[j]] = C[A[j]] - 1
endfor

```

Given that B is an array that can hold up to 8 items. Complete the trace table below and list down what are the values within each cell of array B.

[6]

		*pass 0	Pass 1	Pass 2	Pass 3	Pass 4	Pass 5	
j	A[j]	C[j]	C[j]	C[j]	C[j]	C[j]	C[j]	B[j]
1	3							
2	4							
3	4							
4	1							
5	3							

* where pass 0 contains the new values of array C

- (iii) Explain the role of array C in the second algorithm.

[1]